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IMPACT OF PCA ON LUNG CANCER DATASET CLASSIFICATION: A COMPARITIVE ANALYSIS OF MACHINE LEARNING MODELS

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Abstract— ***Cancer research relies on accurate classification models. Detecting cancer early enhances the chance of a cure. This paper proposes an innovative classification model using enhanced ML (Machine Learning) algorithms for early-stage Lung cancer diagnosis. The study compares the performance of Naive Bayesian, DT (Decision Tree), Ordinal LR (Logistic Regression), and SVM (Support Vector Machine) models before and after PCA (Principal Component Analysis) implementation. The primary objective is to assess how PCA influences classification accuracy in cancer datasets. This study initially evaluates the accuracy rates of four models on a Lung cancer dataset of 250 instances, showing a decrease in accuracy for all four models after PCA implementation, with the Decision Tree model achieving a positive accuracy rate of 94 percent. To enhance accuracy, the dataset volume was increased to 1000 instances, which showed that the Decision Tree model consistently performed exceptionally well, reaching a 100 percent accuracy rate. This highlights the robustness of the DT model and the critical role of model selection in Lung cancer Research.***

Keywords — Cancer classification, Machine learning Algorithms, Principal component analysis, Decision Tree Model, Early-Stage diagnosis

I. INTRODUCTION

Lung cancer is among the most predominant and complex diseases spreading all around the world. The understanding of the disease and detection of the stages of lung cancer improves the patient's recovery and sustainability [1]. This computational technique helps in extracting useful information from a large diverse dataset. This research paper focuses on the application of PCA (Principal Component Analysis) as a powerful tool in the treatment of cancer research. Specifically, Our Investigation objectives are to target on dataset capturing various cancer symptoms to implement some of the ML algorithms like the SVM model, Bayesian model, DT model, and LR model for enhancing the accuracy rate of the dataset, a brief introduction to these models is presented in the below sections. Principal component analysis in the medical field is a very challenging task for dimensionality reduction for datasets [2]. The dataset which is taken for this research is from Kaggle https://www.kaggle.com. The dataset which is initially used in this research paper is about 250 instances and 25 attributes. The stages of our methodology involve meticulous preprocessing of the dataset. The heterogeneous nature of cancer data makes preprocessing imperative tonsure the quality and reliability of the dataset used for subsequent analysis. This encompasses tasks such as handling categorical values, normalizing variables, and addressing outliers to establish a robust foundation for data analysis applications. The research revolves around the judicious selection of tools and frameworks. Python, with its rich ecosystem of libraries, plays a main role in our computational analysis. Various libraries like NumPy, Pandas, and scikit-learn are instrumental in data manipulation and numerical operations and training various models based on mathematical concepts like that of support vector machines, Bayesian models, Decision trees, and logistic regression. Furthermore, the interactive and visual aspects of the analysis are facilitated by tools like Matplotlib and Seaborn. These tools not only contribute to the clarity of our findings but also aid in the interpretation of complex relationships within the dataset. After evaluating the accuracy rate for the lung cancer dataset with the machine learning models, the dataset is increased to 1000 instances for comparing the efficiency of the model's

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impact on the dataset. Comparing algorithms for efficient prediction is a big question and a challenging task in this research [3], this analysis is devoted to building the ML models to help lung cancer prediction and compares the performance of the algorithms [4]. In section 2 we described different research papers that worked with different models and their corresponding outputs. In section 3 the Paper Gives a brief explanation of the Methodology and, in section 4 we describe the experimental findings and analyses of the model. Section 5 describes the conclusion of the paper. LR is applied to determine the relation between one or more independent (predictor) variables along with binary dependent (outcome) variables [5]. SVM belongs to the supervised learning class of ML algorithms, which is based on the concept of hyperplane that separates features into different domains (classes) [2] DT is a supervised learning method and is applied for regression and classification. The DT is an inverted tree structure with a root node at the top followed by branch and leaf nodes [2].

II. LITERATURE REVIEW

Improving patient outcomes requires early lung cancer detection. Recent research in this area has delved the Machine Learning models for the analysis of medical imaging data, particularly Computed Tomography (CT) images [6]. This literature review provides an overview of existing studies and introduces a new research project aimed at improving the understanding of lung cancer detection.

[3], in this paper, the researchers aimed to assess the performance of the DT Classifier, SVM, as well as Naive Bayes Classifier in detecting lung cancer using the dataset of "Lung Cancer Prediction". Results show similar high performance for DT, and NB (Naive Bayes), whereas SVM demonstrates competitive performance. All 3 models exhibit good potential in lung cancer prediction based on recall, accuracy, precision, and F1 values.

[7] , in this paper, the author focused on Early detection and examined the performance of NB, SVM, DT, and LR in predicting lung cancer using CT scans. The study highlights the importance of using classification algorithms to accurately identify lung cancer at different stages.

[8] in the present research, the authors mainly examined the prediction of lung cancer through image processing using various algorithms like NB, DT, Ordinal Logistic Model, KNN (K-Nearest Neighbors), and Random Forest. The objective of the study is to compare accuracy rates and determine the most effective algorithm for predicting lung cancer.

[9] , in this paper, the researchers suggested using the Gaussian NB (GNB) classification algorithm to early detect lung cancer using the dataset from the University of California (UCI). Comparing the results with K-nearest neighbor, SVMs, and the j48 algorithm, the researchers found that GNB is 98% accurate for detecting lung cancer in its early phases, especially with an increased training dataset

[10] , in this paper the researchers highlighted the use of ML in cancer diagnosis, using an ensemble of SVM, NB, and classification trees (C4.5). The study is based on the NCCTG ("North Central Cancer Treatment Group") dataset and highlights the superiority of c4.5 for predicting lung cancer.

[11] , in this paper, the researchers discussed a comprehensive analysis of nine ML algorithms, including LR, KNN Classifier, SVM, and others, that contribute to the prediction and prognosis of lung cancer. The Comprehensive analysis helps clinicians and researchers quickly interpret project results.

A. Limitations of Existing Studies

Existing analyses are based on specific datasets, such as data from the UCI or the NCCTG. This limited data source may not capture the diversity of real-world scenarios, which may affect the generalizability of the algorithms to the wider population.

Some studies lack comprehensive analysis of datasets of different sizes. This limitation may affect the generalizability of the findings to variable-scale data in real-world applications.

The absence of PCA implementation in some studies is a limitation, as PCA is crucial for dimensionality reduction. Evaluating accuracy rates before and after PCA can provide valuable information about the effect of dimensionality reduction on algorithm performance.

III. METHODOLOGY

A. Data Description

The dataset of lung cancer obtained from Kaggle "(https://www.kaggle.com/datasets/thedevastator/cancer-patients-and-air-pollution-a-new-link)", aims to detect lung cancer using various attributes. The tabular dataset comprises 1000 entries and 25 features, with each entry representing a case and each feature corresponding to an attribute for classifying lung cancer into three stages: low, medium, and high. All entries in the dataset are complete, with no missing values, ensuring data integrity. Patient privacy is maintained, and no sensitive information is included. The dataset includes information on lung cancer patients, covering factors like snoring, dry cough, frequent cold, clubbing of fingernails, swallowing difficulty, wheezing, shortness of breath, weight loss, fatigue, coughing of blood, chest pain, passive smoking, smoking, obesity, balanced diet, chronic lung disease, genetic risk, occupational hazards, dust allergy, alcohol use, air pollution exposure, gender, and age. The target variable, labelled "Level," classifies patients into the three stages. To facilitate analysis, preprocessing techniques like label encoding are applied to convert non-numerical values in the target variable into numerical ones. Outliers in the dataset are identified and removed to enhance data accuracy. The patient IDs, initially containing textual values, undergo necessary adjustments. The dataset is split into training (80%) and testing (20%) sets, guaranteeing comprehensive model evaluation. The expansion of the dataset from 250 to 1000 instances is outlined in Table 1.

Table 1-Training and Testing of Dataset

Dataset Size	Training set (80%)	Testing set (20%)
250	200	50

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1000	800	200

B. Principal Component Analysis

An Unsupervised ML algorithm for dimensionality reduction is named PCA ("Principal Component Analysis"). It was originally presented by Mathematician Karl Pearson in 1901 and independently by Hotelling in 1933 [12-14]. A principal component analysis aims to describe the Variance-Covariance structure of the variables by utilizing a small number of linear combinations of the variables in the set. This helps in simplifying the complexity of the data and can be beneficial for various tasks, including visualization and feature extraction. PCA is widely applied for reducing dimensionality in areas such as facial recognition, computer vision, and image compression. Additionally, it plays a crucial role in identifying patterns within high-dimensional data in diverse fields like finance, data mining, bioinformatics, and psychology [15,16,17].

C. Workflow Diagram

The workflow starts with the important step of data collection. This dataset is expected to contain useful information, with each column representing a distinct feature and each row representing an individual data point. Following data collection, it follows a systematic workflow, that starts with the splitting of the dataset into a training phase (80%) as well as a testing phase (20%). In the training phase, the focus is on model development, employing four different models: a Decision Tree, Logistic Regression, a Bayesian Model, and a SVM Model. Each of these models undergoes training using the designated training dataset. The training phase constitutes a crucial step in constructing the research proceeds to the testing phase, where the models are evaluated using the reserved testing dataset. The main aim is to determine the performance of each model and obtain results that illuminate their effectiveness in making predictions on unseen data. This structured approach ensures a comprehensive examination of the Model's capabilities and provides valuable insights into their respective strengths and weaknesses, contributing to the overall findings and conclusions of the research paper.

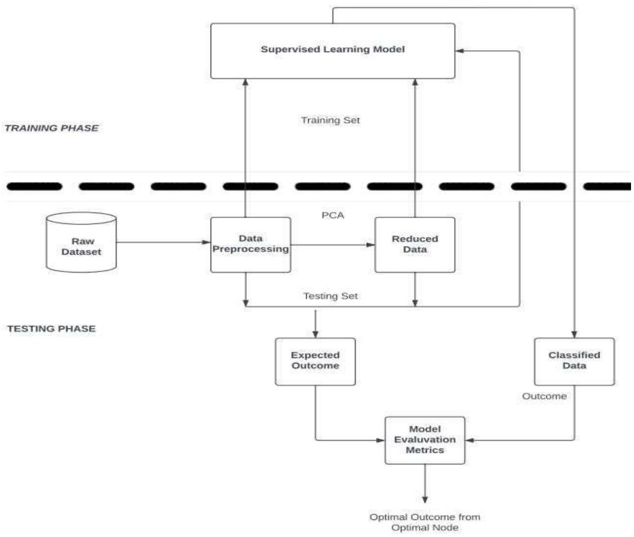

Fig 1. Workflow Diagram

D. Model Description

1. Logistic Regression model

LR model is one of the most popular learning Models. It is used for predicting the categorical dependent variables using a given set of independent variables.

$$log\left(\frac{\_}{1-p}\right) = \beta_0 + \beta_1(x_1) + \cdots + \beta_P(x_P) \qquad (3.1)$$

where β's indicates the regression coefficients x's are independent variables and p represents the probability of the occurrence of the outcome [18]

2. Decision Tree model

DT presents a non-parametric supervised learning method applied for classification as well as regression tasks. One specific method of Decision tree modelling is the Classification and Regression Tree, which involves dividing the data into homogeneous subgroups based on outcome variables. The Decision tree initiation starts at the tree's top, i.e., the root node, where each subsequent node is split into 2 or more subsets. These nodes are determined by the Gini index, a measure of impurity. The Gini index

calculation is as follows: [18]

$$Gini\ Index = 1 - \sum^{i=1} (P_i)^2 \qquad (3.2)$$

3. *Naive Bayes model*

Bayesian theory, the unknown parameter of interest (θ) and the sample data (y) are regarded as random variables with specific distributions. Before incorporating the sample data, a prior distribution is assigned to θ, representing our initial beliefs. The Bayesian formula can be expressed as follows: [19]

$$p\left(\frac{\theta}{y}\right) = \frac{p(\theta)\cdot p\left(\frac{y}{\theta}\right)}{p(y)} \qquad (3.3)$$

The prior distribution of the unknown parameter θ is denoted as P(θ), and the likelihood function derived from the sample data is represented as P(y|θ). The marginal probability is denoted as P(y).

4. Support Vector Machines:

The main purpose of the SVM is to examine hyperplanes that best separate each class from the rest. For multi-class classification (e.g., Low, Medium, High), we have multiple binary classifiers, one for each class. The decision function for class i is:[20]

$$f(X) = \langle w_i, X\rangle + b_i \qquad (3.4)$$

X represents the Feature vector, w_i denotes the weight vector for class i, b_i indicates the bias term for class i.

$< w_i, X >$ signifies the dot product between w_i and X

E. *Confusion Matrix*

[21-25] A confusion matrix is a table utilized in ML to determine the performance of the classification algorithm. It lists the model's predictions and compares them with the actual class labels.

1. Metrics to Evaluate Model Performance

Classification Report: A Visualization tool showcasing four fundamental parameters of a classification model: F1- score,

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Recall, Precision, and Support. These metrics collectively determine the model's accuracy. [21-25]

Accuracy is a metric that measures the model's overall correctness.[21-25]

$$Accuracy = \frac{Tp+TN}{TP+TN+FP+FN} \qquad (3.5)\ [21\text{-}25]$$

Precision: It is determined as the ratio of true positives to the sum of true positives and false positives. [21-25]

$$Precision = \frac{TP}{TP+FP} \qquad (3.6) \quad [21\text{-}25]$$

Recall: It is the ratio of true positives to the total of true positives and false negatives also denoted as True Positive rates or sensitivity . [21-25]

$$Recall = \frac{Tp}{TP+FN} \qquad (3.7) \quad A \quad [21\text{-}25]$$

F1-score: A balanced measure of a model's performance, calculating the harmonic mean of precision and recall. [21-25]

$$F1 = 2 * \frac{Precision*Recall}{Precision+Recall} \qquad (3.8) \quad [21\text{-}25]$$

Support: It represents the count of instances for each class in the dataset

IV. RESULTS AND DISCUSSION

MODEL	WITH 250 INSTANCES		WITH 1000 INSTANCES	
	BEFORE PCA	AFTER PCA	BEFORE PCA	AFTER PCA
REGRESSION	72%	26%	85%	28%
DECISION TREE	94%	74%	100%	81%
NAÏVE BAYESIAN	84%	34%	90%	40%
SVM	38%	38%	41%	40%

Table 2 - Accuracy Rates of Different models

Table 2 describes the accuracy rate of different models implemented on the 250 instances dataset and 1000 instances dataset. From table 2 it is shown that the DT Model performed an accuracy rate of 94% before the implementation of PCA and it consistently performed best after the implementation of PCA with an accuracy of 74%. With the 1000 instances dataset before the implementation of PCA, Decision Tree withstand with the highest accuracy rate of 100% and it consistently performed best after the implementation of PCA with an accuracy rate of 81%. Table 2 describes that the Decision Tree Model performed well among all for the lung cancer dataset The graph given below explains the cumulative variance ratio vs the Number of principal components. From retaining two principal components it results in around 96% of the actual value of the dataset.

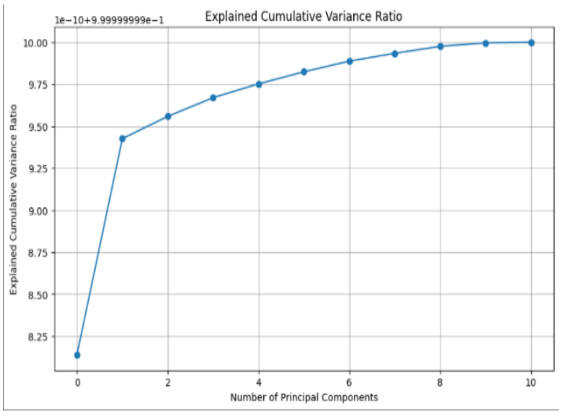

Fig 2 Explained Cumulative Proportional vs principal components retained graph

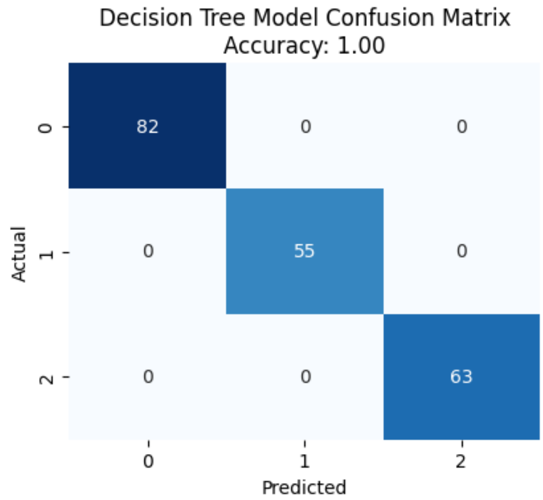

Fig 3 Confusion matrix of Decision Tree Model

Based on the confusion matrix, all 82 records are correctly classified at the zeroth level. At the first level, all 55 records are correctly classified, and at the second level, all 63 records are correctly classified. The confusion matrix of the Decision Tree model proves that it has a high accuracy of 100% compared to other algorithms.

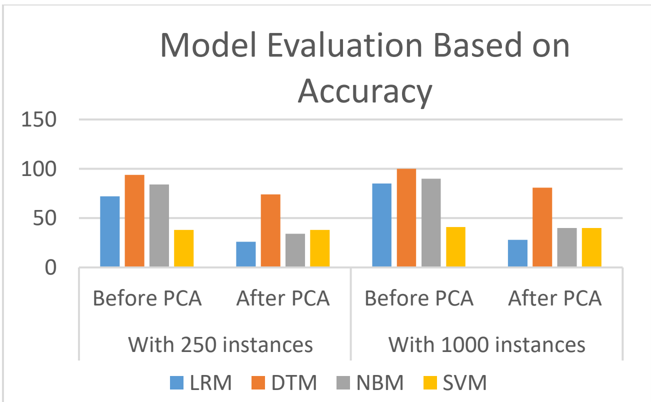

Fig 4 Comparing the Accuracy of Four Different Models

Figure 4 describes that, among the four machine learning algorithms, the Decision Tree model consistently performed best in 250 instances and 1000 instances after applying PCA.

V. CONCLUSION

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In Conclusion, this study investigated how PCA influences the accuracy of classification in Lung cancer datasets, using four Machine learning models. Although the application of PCA led to a decrease in accuracy, this suggests a negative correlation between feature reduction and predictive performance in cancer risk assessment. The Decision Tree model exhibited robust performance, achieving 94% accuracy in a 250-instance dataset and reaching 100% accuracy when scaled to 1000 instances. This highlights the reliability and efficiency of the model in Lung cancer classification even after PCA.

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